GEMSTAR / ARM-FPGA RESEARCH

TASK 15 - FINAL RESEARCH REPORT

Dataset Integrity, Sensor Coverage, Configuration Evidence and Reproducibility

Research phase: Scientific foundation

Status: Archive evidence compiled; calibration and scenario provenance remain under investigation.

Metric	Observed result
OSI file references	49
Unique OSI SHA-256 hashes	38
Unique SensorView recordings	19
Candidate GT/SV pairs	19
Full three-sensor payload recordings	10
Recordings missing sensor payloads	9
Open research issues	7

1. Research objective

Task 15 establishes a reproducible evidence archive for previous GEMSTAR experiments. It preserves file references and integrity hashes, inventories available scenarios, examines SensorView payload coverage, and documents unresolved limitations.

This task does not perform validated object-level sensor fusion, prove calibration correctness, or establish FPGA performance.

2. Experimental data inventory

The dataset manifest records 49 OSI file references with 38 distinct SHA-256 hashes. Repeated hashes indicate identical file content.

3. GT / SensorView pairing

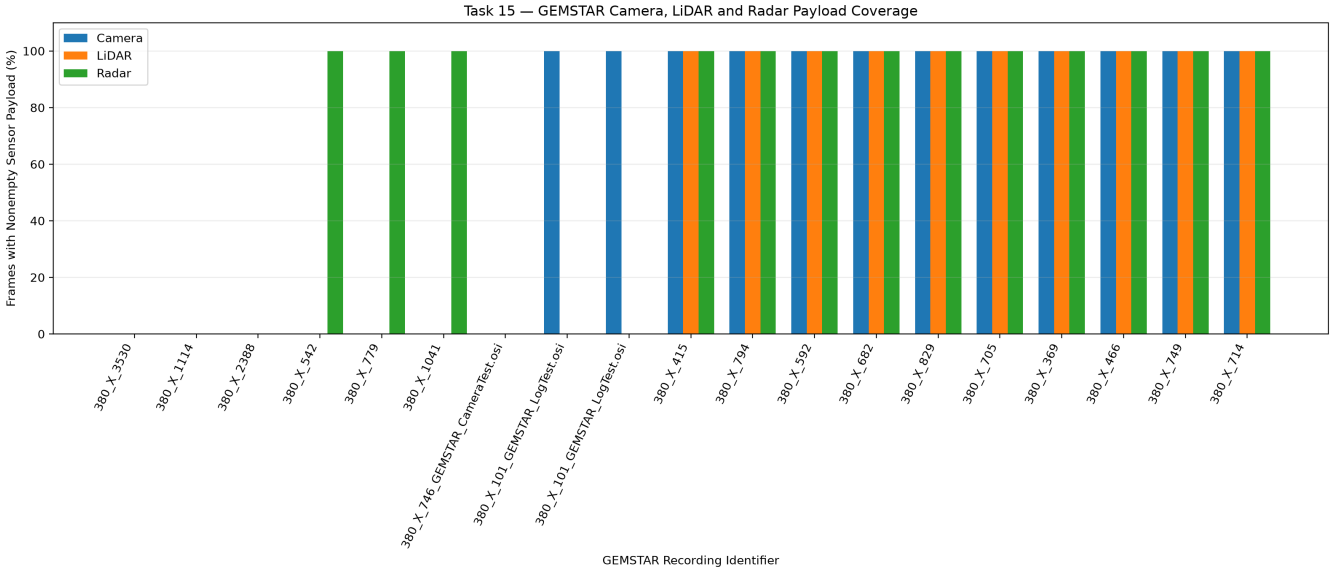
Filename and timestamp analysis produced 19 GT/SensorView candidate pairs. Matching recording identifiers are not conclusive evidence of synchronized sensor measurements. Internal timestamps will be audited in Task 17.

SV recording	GT candidate	Filename delta (s)
121650Z_sv_380_X_3530_GEMSTAR.osi	121650Z_gt_380_X_3530_GEMSTAR.osi	0
121711Z_sv_380_X_1114_GEMSTAR.osi	121711Z_gt_380_X_1114_GEMSTAR.osi	0
123618Z_sv_380_X_2388_GEMSTAR.osi	123618Z_gt_380_X_2388_GEMSTAR.osi	0
125122Z_sv_380_X_542_GEMSTAR.osi	125122Z_gt_380_X_542_GEMSTAR.osi	0
132027Z_sv_380_X_779_GEMSTAR.osi	132027Z_gt_380_X_779_GEMSTAR.osi	0
132427Z_sv_380_X_1041_GEMSTAR.osi	132427Z_gt_380_X_1041_GEMSTAR.osi	0
133514Z_sv_380_X_746_GEMSTAR_Camera	133514Z_gt_380_X_746_GEMSTAR_Camera	0
134242Z_sv_380_X_101_GEMSTAR_LogTes	134242Z_gt_380_X_101_GEMSTAR_LogTes	0
134308Z_sv_380_X_101_GEMSTAR_LogTes	134308Z_gt_380_X_101_GEMSTAR_LogTes	0
142125Z_sv_380_X_415_GEMSTAR.osi	142127Z_gt_380_X_415_GEMSTAR.osi	2
152743Z_sv_380_X_794_GEMSTAR.osi	152743Z_gt_380_X_794_GEMSTAR.osi	0
194512Z_sv_380_X_592_GEMSTAR.osi	194512Z_gt_380_X_592_GEMSTAR.osi	0
204706Z_sv_380_X_682_GEMSTAR.osi	204706Z_gt_380_X_682_GEMSTAR.osi	0
205013Z_sv_380_X_829_GEMSTAR.osi	205014Z_gt_380_X_829_GEMSTAR.osi	1
222053Z_sv_380_X_705_GEMSTAR.osi	222054Z_gt_380_X_705_GEMSTAR.osi	1
223857Z_sv_380_X_369_GEMSTAR.osi	223857Z_gt_380_X_369_GEMSTAR.osi	0
224313Z_sv_380_X_466_GEMSTAR.osi	224313Z_gt_380_X_466_GEMSTAR.osi	0
210045Z_sv_380_X_749_GEMSTAR.osi	210046Z_gt_380_X_749_GEMSTAR.osi	1
210709Z_sv_380_X_714_GEMSTAR.osi	210710Z_gt_380_X_714_GEMSTAR.osi	1

4. Sensor payload coverage

All 19 unique SensorView recordings were parsed without reported parser errors. 10 recordings had nonempty Camera image data, LiDAR reflection records and Radar reflection records in every parsed SensorView message. The other 9 recordings lacked at least one populated sensor stream.

Payload coverage is not equivalent to positive/valid returns, calibration accuracy, temporal synchronization or verified object associations.



Recording ID	Frames	Camera	LiDAR	Radar
3530_GEMSTAR	3530	0	0	0
1114_GEMSTAR	1114	0	0	0
2388_GEMSTAR	2388	0	0	0
542_GEMSTAR	542	0	0	542
779_GEMSTAR	779	0	0	779
1041_GEMSTAR	1041	0	0	1041
746_GEMSTAR_CameraTe	746	0	0	0
101_GEMSTAR_LogTest	101	101	0	0
101_GEMSTAR_LogTest	101	101	0	0
415_GEMSTAR	415	415	415	415
794_GEMSTAR	794	794	794	794
592_GEMSTAR	592	592	592	592
682_GEMSTAR	682	682	682	682
829_GEMSTAR	829	829	829	829
705_GEMSTAR	705	705	705	705
369_GEMSTAR	369	369	369	369

Recording ID	Frames	Camera	LiDAR	Radar
466_GEMSTAR	466	466	466	466
749_GEMSTAR	749	749	749	749
714_GEMSTAR	714	714	714	714

5. Scenario inventory

Scenario folder names are treated as provisional evidence. Historical recording-specific weather parameters have not been conclusively reconstructed.

Folder	GT	SV	Verification
baseline_clear	1	1	NO
baseline_fog	0	1	NO
baseline_fog_working_radar	1	1	NO
fog_mor100	1	1	NO
fog_mor20	0	0	NO
fog_mor50	0	0	NO
fog_postfix_validation	1	1	NO
task05_fog_test	1	1	NO

6. Configuration evidence

Six configuration snapshots were archived and hashed. Camera and LiDAR mount parameters were identified; the Radar mounting convention needs further investigation. Logger inputs differed between inspected configuration versions.

Configuration file	SHA-256 prefix
DefaultGemstar.ini	585CEE7593ABA0F3
DefaultGemstar_backup.ini	9474EF97173C07E6
DefaultGemstar_before_radar_test.ini	3655950C924B173F
DrivingPractiseAreaBrunswickMap.sumocfg	7392C3F5FA1E8A09
FeatureDefaultsWithSumoInput.ini	F2F975FFC64AB7B6
GemstarWeatherTest.ini	01BD40945A0ABD66

A configuration snapshot shows what a file contains, not necessarily which settings were active when an earlier recording was generated.

7. Open research issues

ID	Finding	Next task
T15-001	9 of 19 unique recordings lack at least one populated sensor stream	17-20
T15-002	fog_mor20 and fog_mor50 research folders contain no recordings	19 and 25
T15-003	Recording 415 GT counterpart exists in GEMSTAR Output but not in baseline_fog folder	17
T15-004	Radar Mount setting commented out in inspected configuration snapshots	17 and 20
T15-005	Historical configuration for each recording has not been conclusively identified	17 and 25
T15-006	Full payload coverage does not establish valid returns, calibration, synchronization or object association	17-24
T15-007	Earlier Task 14 Camera-Radar association provisional and target LiDAR association unconfirmed	17-21

8. Interpretation and scientific limitations

- The ten full-coverage recordings are candidates for subsequent three-sensor investigations, not yet validated fusion datasets.

- The nine incomplete recordings are retained as evidence of earlier development and logging configurations.

- The fog_mor20 and fog_mor50 dataset directories have no recordings in the inspected research archive.

- Radar geometry, historical configuration provenance, sensor synchronization and target association remain unresolved.

9. Next research stages

Task	Next objective
16	Literature matrix and testable research hypotheses
17	Sensor timestamps, coordinate systems and calibration
18	Camera features and multiframe observability
19	LiDAR filtering, geometry and quality
20	Radar measurement semantics and quality
21	Validated cross-sensor object association

10. Reproducibility package

Associated research artifacts are stored in the Task_15_Final_Research folder: 01_Graphs, 02_CSV_Data, 03_Configurations, 04_Code, 05_Raw_Data_References, 06_Research_Findings and 07_Research_Report.

The original OSI files remain at their existing paths. References and SHA-256 hashes are preserved rather than duplicating multi-gigabyte recordings.